

PRESENTATION ON

BOOTSTRAP 4 AND 5.X

BOOTSTRAP OVERVIEW

What is Bootstrap?

- Free and open-source CSS framework
- Provides responsive design components
- Built with HTML, CSS, and JS
- Speeds up front-end development
- Used in millions of websites

BOOTSTRAP 4 VS 5

Feature	Bootstrap 4	Bootstrap 5
jQuery Dependency	Required for components	Removed completely
Grid System	5 breakpoints (xs to xl)	6 breakpoints (added xxl)
Form Controls	Basic styling, some customization	Improved layout, better validation
Custom CSS Variables	Limited use	Widely used for easier theming
Icons	Not included	Bootstrap Icons (separate package)
Utilities API	Static utility classes	Customizable utility generator
RTL Support	Not supported	Fully supports RTL (Right to Left)

GRID SYSTEM

- Based on 12-column layout
- Responsive breakpoints: xs, sm, md, lg, xl
- Use .container, .row, .col-*
- Nesting is supported

```
<div class="container">
  <div class="row">
    <div class="col-md-6 bg-primary text-white">Column 1</div>
    <div class="col-md-6 bg-success text-white">Column 2</div>
  </div>
</div>
```

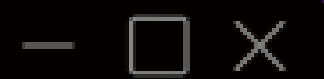

COMMON COMPONENTS

- Buttons, Alerts, Badges
- Cards and Navbars
- Responsive Navigation
- Easy customization

```
<div class="card" style="width: 18rem;">
  <div class="card-body">
    <h5 class="card-title">Example Card <span class="badge bg-secondary">New</span></h5>
    <p class="card-text">This is a simple card with Bootstrap styling.</p>
    <a href="#" class="btn btn-primary">Click Me</a>
  </div>
</div>
```

NAVBAR AND PAGINATION

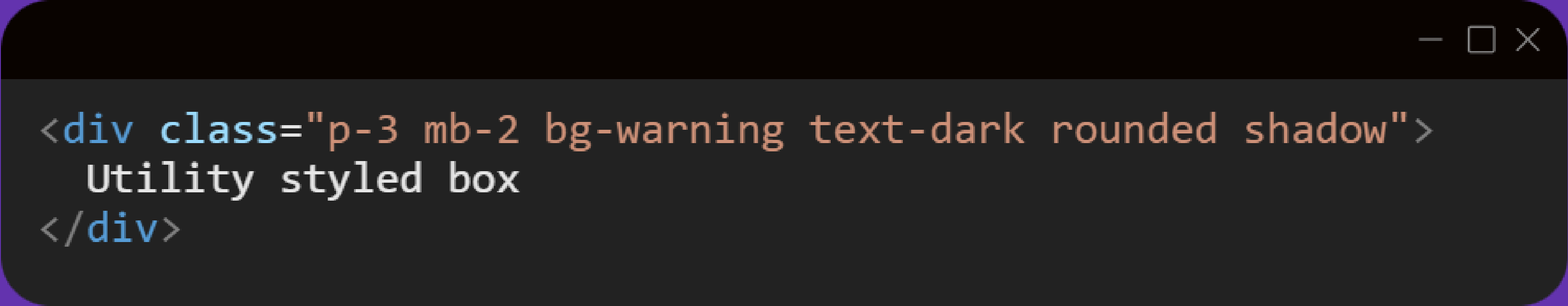
- Responsive Navbar collapses on small screens
- Built-in support for toggle, dropdowns
- Pagination for navigation between pages



```
<nav class="navbar navbar-expand-lg navbar-light bg-light">
  <a class="navbar-brand" href="#">Brand</a>
  <button class="navbar-toggler" type="button" data-bs-toggle="collapse" data-bs-target="#navbarNav">
    <span class="navbar-toggler-icon"></span>
  </button>
  <div class="collapse navbar-collapse" id="navbarNav">
    <ul class="navbar-nav">
      <li class="nav-item"><a class="nav-link active" href="#">Home</a></li>
      <li class="nav-item"><a class="nav-link" href="#">Features</a></li>
    </ul>
  </div>
</nav>
```

UTILITY CLASSES

- m- / p- classes for margin and padding
- text-* for text color
- bg-* for background
- border, rounded, shadow for styling
- Visibility: d-none, d-block, etc.



```
<div class="p-3 mb-2 bg-warning text-dark rounded shadow">  
  Utility styled box  
</div>
```

FLEXBOX UTILITIES

- Use d-flex to make an element flex container
- Align items with justify-content-*, align-items-*
- Useful for layouts, spacing, alignment

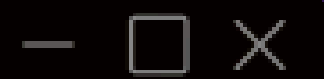
```

<div class="d-flex justify-content-between bg-light p-3">
  <div>Left</div>
  <div>Right</div>
</div>

```


FORMS

- Use form-control, form-group, form-label
- Supports validation styles
- Bootstrap 5 has better spacing and input styling



```
<form>
  <div class="mb-3">
    <label class="form-label">Email address</label>
    <input type="email" class="form-control" placeholder="name@example.com">
  </div>
  <button type="submit" class="btn btn-success">Submit</button>
</form>
```

MODALS AND CAROUSELS

- Modals for pop-ups (login, alerts)
- Carousels for image sliders
- Controlled using data attributes or JS


```
<!-- Button trigger -->
<button class="btn btn-primary" data-bs-toggle="modal" data-bs-target="#myModal">Open Modal</button>

<!-- Modal -->
<div class="modal fade" id="myModal" tabindex="-1">
  <div class="modal-dialog">
    <div class="modal-content">
      <div class="modal-header">
        <h5 class="modal-title">Modal title</h5>
        <button class="btn-close" data-bs-dismiss="modal"></button>
      </div>
      <div class="modal-body">Hello, this is a Bootstrap Modal!</div>
    </div>
  </div>
</div>
```

THANK YOU !